

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 39189

LM PHYSICS OF RENEWABLE ENERGY

SUMMER EXAMINATION 2024

Time Allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.
Answer ***all four*** questions from this Section.

Section 2 consists of three questions and carries 60% of the marks.
Answer ***two*** questions from this Section. If you answer more than two questions,
credit will only be given for the best two answers.

The approximate allocation of marks to each part
of a question is shown in brackets [].

Calculators may be used in this examination but must not be used to store text.
Calculators with the ability to store text should have their memories deleted prior to
the start of the examination.

A formula sheet and a table of physical constants and units that may be required will
be found at the end of this question paper.

SECTION 1

Answer **all four** questions from this Section.

1. (a) Describe the relevance of features of Fig. (1) to understanding the greenhouse effect. [4]
- (b) Describe the features of relevance in Fig. (1) to proposed space-based photovoltaic collection. [2]
- (c) How large in area would a structure in space need to be to provide 40GW for the UK, assuming 1300 Wm^{-2} solar flux outside atmosphere, 30% efficiency and neglecting transmission loss in other aspects of the system? [2]
- (d) What are the advantages of space-based collection versus terrestrial photovoltaic power? [2]

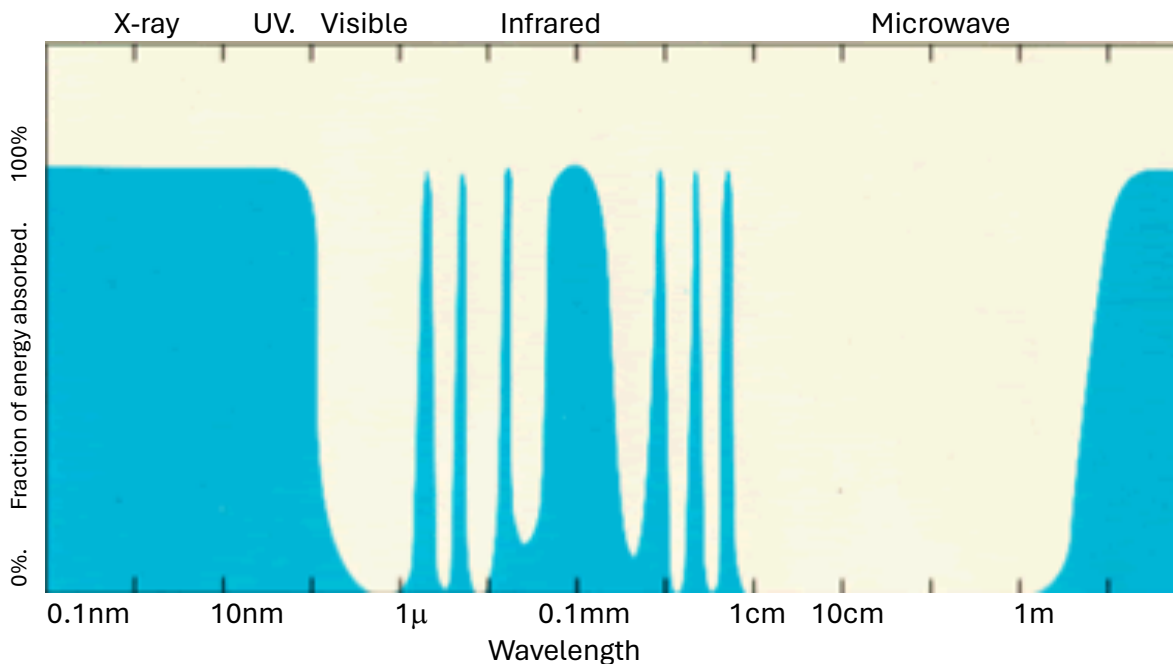


Figure 1: Atmospheric transmission of electromagnetic radiation in the visible (lefthand side of figure); the infrared in the middle; and the microwave region. Blue regions indicate absorption – see vertical axis. 2.5 GHz has a wavelength of 12 cm. Credit: Frederick.

ANY CALCULATOR

2. Consider two wind turbines, one with two blades and one with three blades. In both cases they are symmetrical – with an angle of π between the blades in the first case and $2\pi/3$ in the second. Note $\cos(2\pi/3) = -1/2$.

To align the wind turbine with the direction of the wind, which may change, it must rotate around the *vertical* axis, while it is spinning, with a uniform angular velocity, Ω , around a *horizontal* axis.

Calculate the moment of inertia around the *vertical* axis in each case as a *function of time*. Use an idealised description of the turbine blades as having all their mass, m , at the end of the blade of length ℓ .

Comment on any practical consequences of your result.

[10]

3. A depletion layer at **zero-temperature** has its normal in the z -direction, with (ionised) donors between $-\ell \leq z < 0$ of concentration n , and (ionised) acceptors in the region $0 < z \leq \ell$, also of concentration n . The donors and acceptors outside the depletion layer are assumed neutral.

(a) Deduce the electric field at $z = 0$, assuming there is no applied external field.

[6]

(b) Draw the band edges as a function of z , indicating regions of both neutral and ionised donors and acceptors.

[4]

ANY CALCULATOR

4. In a cold winter, the UK electrical power demand can have a daily average 40 GW.

- (a) If there were a windless dark fortnight, how much energy, E , in Wh would need to be stored if there were no other sources of power? **[2]**
- (b) Compare storage of E , in terms of volume, using: (i) AA batteries (each of which contains roughly 4Wh of energy), which are 5 cm long with a radius of 0.7 cm; (ii) in the electric field of a capacitor filled with Barium Titanate, which has a relative dielectric permittivity $\varepsilon = 1200$, and a maximum (breakdown) field of 10^8 Vm^{-1} . **[6]**
- (c) If the batteries or capacitor were used as an electrical supply, is there any particular advantage in using one of these modes of storage? **[2]**

SECTION 2

Answer **two** questions from this Section. If you answer more than two questions, credit will only be given for the best two answers.

5. Consider a spherical planet with radius a , with no atmosphere, a distance R from the star where $a \ll R$. At R the flux of energy from the star is I_s per unit area per unit time. The planet has no internal source of heat and radiates as a black body at temperature T_p , with emissivity $e_p = 1$

- (a) Show that the steady state temperature of the planet is [8]

$$T_{p,0} = \left(\frac{I_s}{4\sigma} \right)^{1/4}.$$

Now let there be N thin layers of gas around the planet. These do not absorb at the frequencies of the radiation from the star, but absorb at the frequencies radiated from the planetary surface.

The total thickness is much less than the radius of the planet, so we may consider the surface of the planet to be flat and the layers of gas to be parallel to the surface. The flux arriving at the surface is now αI_s , where $0 < \alpha \leq 1$, allowing for the albedo of the atmosphere.

Let the temperature of the layers be T_m , where $1 \leq m \leq N$ and the temperature of the surface of the planet be $T_{p,N}$. Assume steady state, with only radiative transfer of energy (i.e. disregarding convection and conduction of heat).

- (b) If $N = 1$, show $T_{p,1} = 2^{1/4} T_{p,0}$. [8]

- (c) If $N = 2$, show $T_{p,2} = 3^{1/4} T_{p,0}$. [4]

- (d) Write down the balance of emission and absorption of radiation by the m th layer (where $1 < m < N$) of the atmosphere and the result for the $m - 1$ th layer. Hence show that

$$T_{n-m}^4 = \frac{m+1}{m} T_{n-(m-1)}^4.$$

[6]

- (e) Deduce the surface temperature for N layers, $T_{p,N}$. [4]

ANY CALCULATOR

6. Consider the trailing vortices in the wake of a wind turbine with three symmetrically arranged blades, of length Λ . The wind velocity a long way in front of the turbine (upwind) is U . The turbine is rotating with angular velocity Ω . Assume there is no interaction between the vortices.

- (a) Sketch the vortices explaining why they are helical (see formula sheet for definition of terminology) and writing down an expression for the pitch of each of the three helices in terms of their velocity, U_V . Do not consider the starting vortices. [10]

On the downwind side, denote the velocity on the axis of the helix to be fU and the velocity outside the helix to be U . Assume the vortices move with the average of these velocities, i.e. $U_V = \frac{1}{2}U(1 + f)$.

- (b) Use Stokes' theorem to show the circulation of each trailing vortex, Γ , obeys [8]

$$\Gamma = \frac{2\pi}{3\Omega}(1 - f^2)U^2.$$

The equation of motion of an incompressible fluid (of density ρ_0), including the effect of viscosity, is

$$\frac{\partial \mathbf{u}}{\partial t}(\mathbf{r}, t) + \frac{1}{2}\nabla(\mathbf{u}^2) - \mathbf{u} \times (\nabla \times \mathbf{u}) = -\nabla \frac{p}{\rho_0} - \nabla \mathcal{V} + \nu \nabla^2 \mathbf{u}, \quad (1)$$

where ν is the kinematic viscosity. The vorticity is defined to be $\boldsymbol{\omega} = \nabla \times \mathbf{u}$

- (c) By invoking a vector calculus identity, show that $\nabla \cdot \boldsymbol{\omega} = 0$. What is the value of $\nabla \cdot \mathbf{u}$ for an incompressible fluid? [1]

- (d) Take the curl of Eq. (1), for an incompressible fluid, to obtain the equation of motion for $\boldsymbol{\omega}$, noting the similarity with the diffusion equation. [The identity [4]

$$\begin{aligned} \nabla \times [\mathbf{u} \times (\nabla \times \mathbf{u})] &= \nabla \times (\mathbf{u} \times \boldsymbol{\omega}) \\ &= -(\mathbf{u} \cdot \nabla)\boldsymbol{\omega} - \boldsymbol{\omega}(\nabla \cdot \mathbf{u}) + (\boldsymbol{\omega} \cdot \nabla)\mathbf{u} + \mathbf{u}(\nabla \cdot \boldsymbol{\omega}), \end{aligned}$$

maybe useful.]

- (e) Consider the case where $\boldsymbol{\omega} \cdot \nabla \mathbf{u} = 0$. By using dimensional analysis, combined with physical arguments, determine a length scale upon which the wake of the wind turbine relaxes. (And hence a sensible minimum separation of the turbines in the direction of the prevailing wind.) [7]

7. (a) Explain the features in the graph of the absorption length for electromagnetic waves in Si (right hand figure in Fig. (2)) by reference to the main features (not details) of the band structure of Si (left hand figure in Fig. (2)). [8]

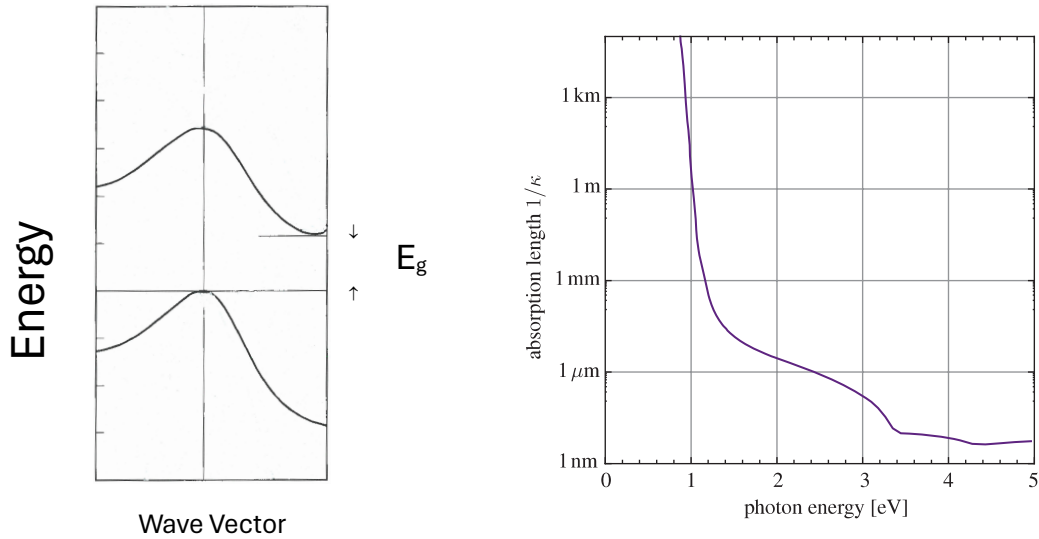


Figure 2: **Left:** Band structure of Si. **Right:** Absorption length for electromagnetic waves in Si as a function of frequency. Credit: Jaffe and Taylor.

- (b) Sketch the intensity of solar radiation in the visible part of the spectrum and, on the same figure, the fraction of that power **usefully** absorbed by a Si sample.
- (c) Explain the form of the curve of the fractional power usefully absorbed from part (b) and explain why the efficiency of the collection of solar power is: [8]

$$\eta_{\text{coll}} = \frac{E_g \int_{E_g}^{\infty} \frac{dE E^2}{e^{\beta E} - 1}}{\int_0^{\infty} \frac{dE E^3}{e^{\beta E} - 1}}, \quad (2)$$

where $\beta = 1/k_B T_{\text{Sun}}$.

Consider a device with two pn-junctions, with **indirect** band gaps $E_g^a < E_g^b$. It is desired to recover more power than a single pn-junction. Both junctions are thicker than their visible frequency absorption lengths.

- (d) Which junction should the sunlight first encounter? Why? [2]
- (e) Construct the counterpart to Eq. (2) for the two junction case. [5]
- (f) Consider a device where the band gap varied spatially continuously from $E_g^{\min}$ to $E_g^{\max}$, in the appropriate orientation to maximise power. Write down the counterpart to Eq. (2) assuming absorption length varies slowly through the sample. [7]

Formula Sheet

Stefan-Boltzmann law

The flux, F , of energy per unit area per unit time from a body at temperature T with emissivity $0 \leq e \leq 1$ is

$$F = e\sigma T^4 ,$$

where σ is the Stefan-Boltzmann constant.

Solar flux

At the distance of the Earth from the sun, the radiative flux of energy from the Sun is 1100 Wm^{-2} at sea level.

Energy density of an electric field in a dielectric

$$\mathcal{E} = \frac{1}{2}\epsilon\epsilon_0\mathbf{E}^2 .$$

Stokes' theorem

For a vector field $\mathbf{v}(\mathbf{r})$, with an open surface S with boundary C .

$$\int_C d\mathbf{r} \cdot \mathbf{v} = \int_S d\mathbf{S} \cdot \nabla \times \mathbf{v} .$$

Gauss' theorem

For a vector field $\mathbf{v}(\mathbf{r})$ with a closed surface S enclosing a volume V :

$$\int_S \mathbf{v} \cdot d\mathbf{S} = \int_V dV \nabla \cdot \mathbf{v} .$$

Typical wavelength of light in the visible spectrum $\simeq 500 \text{ nm}$

Indirect band gap of Si $\simeq 1.1 \text{ eV}$

Thermal energy room temperature $k_B T \simeq 25 \text{ meV}$ at room temperature $\simeq 300 \text{ K}$.

Definition of a helix

$$\mathbf{r}(\theta) = a (\cos \theta \hat{\mathbf{x}} + \sin \theta \hat{\mathbf{y}}) + \frac{b\theta}{2\pi} \hat{\mathbf{z}} ,$$

where a is the *radius* of the helix and b is the *pitch* of the helix. The *axis* of the helix is the z -axis.

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[N.B. 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.